Red-Blue Hamiltonian Tomography via the Operator Lightcone Wavemap

Paolo D'Alberto
paolo.dalberto@amd.com
Advanced Micro Devices, Inc.

Abstract

We introduce a model-free method for Hamiltonian tomography based on the operator spreading wavefront. Given two tensor network simulations of the same quantum lattice (one with the nominal Hamiltonian and one with one or more perturbed bonds), the signed cross-entropy of the non-identity Pauli weight field $\mathcal{L}(v,t) = -\ln(n_{\text{pert}}/n_{\text{base}})$ identifies each perturbed bond by its causal signature: a red-blue sign flip at the two endpoints, visible from the first wavefront crossing. Combining measurements from p strategically placed observation centers via an ensemble log-likelihood sum, a single broken bond is identified with an 82-fold margin over the next candidate on a 7×7 lattice. Extending to realistic coupling degradation ($\delta J = -0.20$) and up to 8 simultaneous faults, the integrated log-ratio $\Lambda(v) = \ln(\mathcal{B}(v)/\mathcal{N}(v))$ achieves a per-bond detection rate above 90% in a single seismic pass using only 6 simulations. An iterative calibration loop reduces multi-bond detection to a sequence of single-bond problems, each solved at 100% accuracy. The method requires no model assumptions and no knowledge of the perturbation type or magnitude; the detection is deliberately blind.

1 Introduction

Identifying miscalibrated or anomalous couplings in a quantum processor is a central challenge in quantum hardware characterization. Standard approaches require explicit Hamiltonian learning: fitting a parameterized coupling model to measurement data. This scales poorly with system size and depends on model assumptions about the noise and interaction structure.

We propose a different approach based on a clear separation of roles. The *simulator* knows the nominal Hamiltonian and produces two Pauli weight fields: $n_{\text{base}}(v,t)$ from the nominal evolution, and $n_{\text{pert}}(v,t)$ from an evolution where one bond has been silently modified. The *detection algorithm* receives only these two arrays of numbers and must identify the modified bond from their ratio alone, with no knowledge of the Hamiltonian, the perturbation type, or its magnitude. The detection is deliberately blind.

A perturbed bond creates a causal signature in the wavefront: a sign flip in the cross-entropy field $\mathcal{L}(v) = -\ln(n_{\text{pert}}/n_{\text{base}})$ between its two endpoints that propagates no further. The broken bond is exactly where the wavefront sign changes, identifiable from the observables alone.

The method is inspired by seismic tomography: place sensors (observation centers) at strategic positions, emit a wave (initialize a local operator), and read the arrival pattern. The perturbed bond is the anomaly in the medium.

The red-blue color scheme is a deliberate reference to the red-blue pebble game of Hong and Kung [2], which characterizes I/O complexity as a bottleneck in computational graphs. Here the roles are analogous: red sites (upstream, wave blocked) and blue sites (downstream, wave starved)

identify the bond that limits operator propagation, exactly as red and blue pebbles identify the I/O bottleneck in a computation.

2 Setup

We simulate the kicked isotropic Heisenberg model on a 7×7 rectangular lattice (49 sites, 84 bonds) using a GPU-accelerated belief propagation tensor network simulator with Trotter step $\varepsilon = 0.10$, bond dimension $\chi = 50$, and truncation cutoff $\delta = 10^{-3}$. The non-identity Pauli weight at site v and cycle t is:

$$n(v,t) = \sum_{P:\, P_v \neq I} c_P(t)^2 \tag{1}$$

where $O_c(t) = \sum_P c_P(t) P$ is the evolved operator initialized at center c. By normalization $\sum_P c_P^2 = 1$, $n(v,t) \in [0,1]$ is a probability.

To simulate miscalibrated bonds, we weaken their couplings by δJ, with each bond potentially taking a different value. In the experiments below we use $\delta J = -0.20$ ($J = 0.80$, a 20% degradation) as a representative choice; the detection method does not depend on knowing δJ in advance and works for any coupling deviation. The Pauli transfer matrix entries scale as $\cos(J\varepsilon)$, attenuating operator propagation without blocking it entirely. For the multi-bond experiments in Section 10, k bonds are selected uniformly at random and perturbed simultaneously.

We place $p = 6$ observation centers using a greedy geodesic bond cover: the minimum set of sites such that every bond is within $d_{\max} = 2$ hops of at least one center. For the 7×7 lattice this gives centers at sites $\{6, 10, 25, 30, 32, 34\}$, covering all 84 bonds.

3 Signal Vocabulary and Measurement Definitions

All definitions are expressed in terms of two observables per site per cycle: $n_{\text{base}}(v,t)$ and $n_{\text{pert}}(v,t)$, the non-identity Pauli weights under the nominal and perturbed Hamiltonian respectively. The BFS frontier crossing time $t_0(v) = \min\{t : n_{\text{base}}(v,t) > \varepsilon_{\text{th}}\}$ defines a site-specific reference time; it marks the physical arrival of the wavefront at v, set by the Lieb-Robinson lightcone, not by the numerical cutoff.

At each site v in the 3-cycle window $[t_0(v) \pm 1]$ we observe one of three frontier patterns. A site is **White** [W] when $n_{\text{base}}(v, t_0) \leq \varepsilon_{\text{th}}$: the baseline wave has not yet arrived and no information is available. A site is **Blue** [B] when $n_{\text{pert}}(v, t_0) > n_{\text{base}}(v, t_0) > \varepsilon_{\text{th}}$: the perturbed wave accumulates at v because drainage is blocked by a broken bond downstream; the site is a *wall face*. A site is **Red** [R] when $n_{\text{pert}}(v, t_0) < n_{\text{base}}(v, t_0)$ with $n_{\text{base}}(v, t_0) > \varepsilon_{\text{th}}$: the perturbed wave is attenuated because a broken bond on the path from center blocked transmission; the site is in the *shadow*.

For a bond (u, v) with u on the center-side and v on the shadow-side, these patterns combine into three observable signatures. When both endpoints are white [W, W, W] the bond has not yet been reached or is healthy. When the wall face is detected but the shadow is not yet visible [W, B, W], one more cycle is needed. The **broken bond signature** [W, B, R] is the primary target: u appears blue at cycle t_0 and v appears red at cycle $t_0 + 1$, identifying the bond between them as anomalous. This pattern rests on a single assumption: both waves reach u at the same frontier time t_0, so any amplitude difference at v is attributable to the bond, not to a timing offset. The temporal order (blue before red by one cycle) is then universal, following from the BFS causal structure: u is reached before v, so the wall accumulates before the shadow appears.

The signed cross-entropy evaluated in the frontier window is:

$$\mathcal{L}(v_0,\,[t_0 \pm 1]) = -\ln \frac{n_{\text{pert}}(v_0,\,t_0)}{n_{\text{base}}(v_0,\,t_0)} \tag{2}$$

This is a *cross-correlation fixed in space*: both waves evaluated at the same site v_0 at the same frontier moment t_0, with the window $[t_0 \pm 1]$ providing context (white before, red after). Positive $\mathcal{L}$ identifies red (shadow), negative identifies blue (wall face). The broken bond is the edge between consecutive blue and red sites in the BFS ordering.

A complementary observable measures the wave *through* a bond rather than *at* a site. Consider the bond (v_0, u_0) in the BFS path from the center. The baseline reaches v_0 at frontier time t_0 and u_0 at $t_1 = t_0 + 1$. The *baseline transfer ratio* is:

$$T_{\text{base}}(v_0 \to u_0) = \frac{n_{\text{base}}(u_0,\,t_1)}{n_{\text{base}}(v_0,\,t_0)} \tag{3}$$

This measures how much of the wave at v_0 propagates through the bond to u_0 in one BFS step, directly encoding the gate's coupling strength J. For the perturbed run (with potentially delayed frontier times t_0^p, t_1^p):

$$T_{\text{pert}}(v_0 \to u_0) = \frac{n_{\text{pert}}(u_0,\,t_1^p)}{n_{\text{pert}}(v_0,\,t_0^p)} \tag{4}$$

The ratio $T_{\text{base}}/T_{\text{pert}} > 1$ signals a broken bond: less weight crosses the bond in the perturbed run.

Two important caveats apply. First, there is a path distortion. The perturbed wave arriving at v_0 has already traversed all upstream bonds. If any of those are also broken, $n_{\text{pert}}(v_0, t_0^p)$ is already weakened, not by bond (v_0, u_0) but by upstream damage. The ratio $T_{\text{base}}/T_{\text{pert}}$ then conflates the upstream distortion with the target bond signal. The ratio is not a pure measurement of bond (v_0, u_0). Second, there is a measurable Sensitivity. problem. The perturbed weight $n_{\text{pert}}(v_0, t_0^p)$ may be close to ε_{th}, the threshold, making the ratio numerically unreliable. The local cross-correlation $\mathcal{L}(v_0, [t_0 \pm 1])$ avoids both caveats by using the baseline as the reference; it measures the *relative* change at the frontier moment, not the absolute transfer.

4 The Cross-Entropy Wavemap Observable

The previous section introduced the signal vocabulary for a wave with a single origin: one center fires, the wavefront spreads, and the red-blue sign flip at the bond endpoints is the detection signal. Here we extend to the case where p centers fire independently, each producing its own wavemap. The key quantities are defined per-center, then combined into an ensemble.

For each center k and site v, we evaluate the signed cross-entropy in a 3-cycle window $[t_0 \pm 1]$ centered on the BFS frontier crossing $t_0(v) = \min\{t : n_{\text{base}}^k(v,t) > \varepsilon_{\text{th}}\}$:

$$\mathcal{L}_k(v,\,[t_0 \pm 1]) = -\ln \frac{n_{\text{pert}}^k(v,\,t_0(v))}{n_{\text{base}}^k(v,\,t_0(v))} \tag{5}$$

This is a *cross-correlation fixed in space*: both waves are evaluated at the same site v and the same frontier moment $t_0(v)$, defined by the baseline. The window $[t_0 - 1, t_0, t_0 + 1]$ captures three states — before arrival (WHITE), wall-face signal (BLUE or WHITE), and shadow signal (RED) — and $t_0(v)$ is the moment of maximum signal and minimum χ-saturation noise.

Consider a bond (u, v) with u closer to center k. At time t_0, the baseline wave has just crossed the bond and $n_{\text{base}}(u, t_0) \approx n_{\text{pert}}(u, t_0)$; both waves have reached u with equal amplitude. One cycle

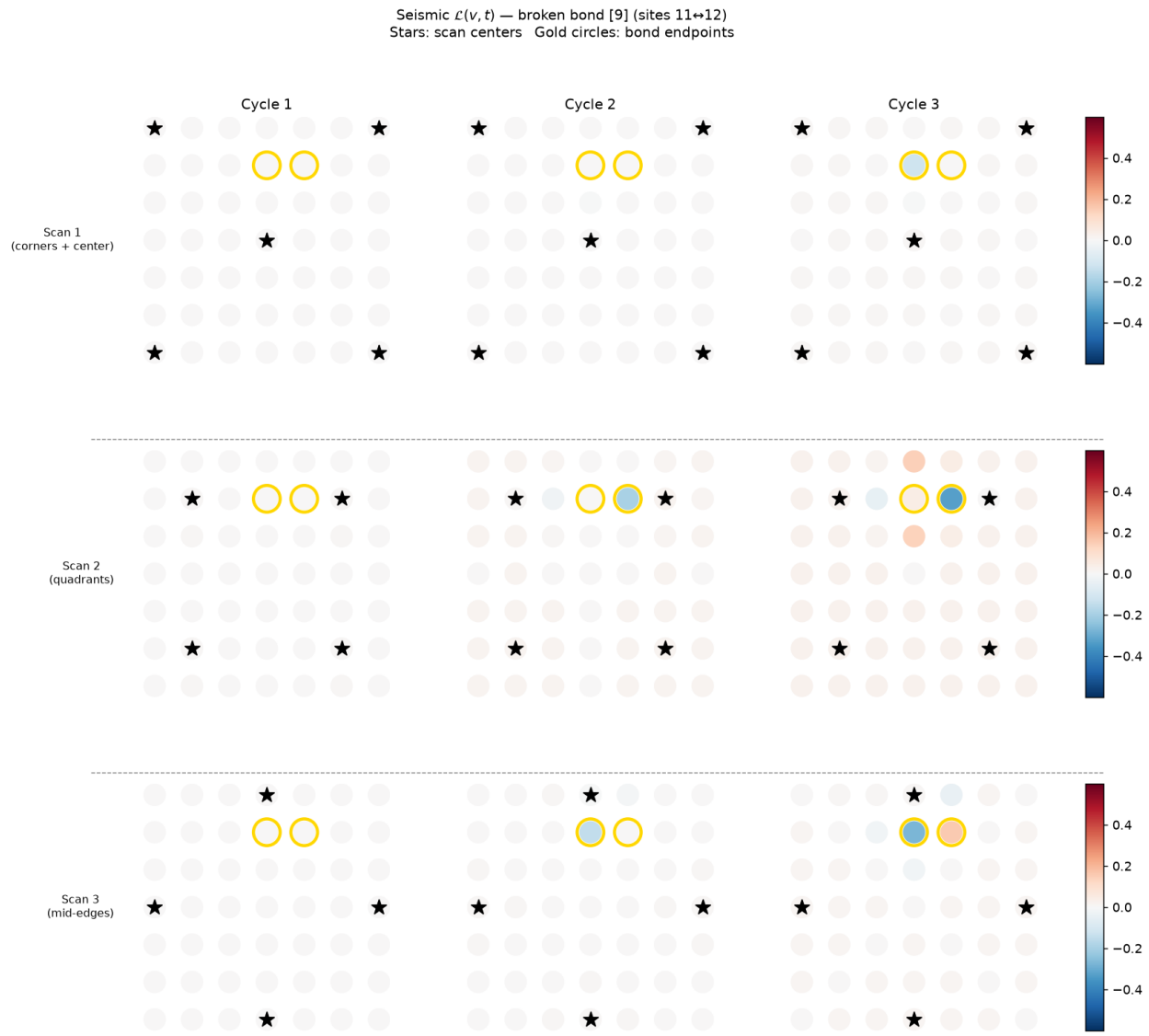


Figure 1: Signed cross-entropy $\mathcal{L}(v,t)$ for a single broken bond (bond 9, sites 11↔12, $\delta J = -0.20$), rect 7×7, $\chi = 50$, 3 cycles. Rows show the three seismic scans: corners+center, quadrants, mid-edges. Columns show cycles 1–3. **Cycle 1**: no signal in any scan. **Cycle 2**: yellow (BLUE) accumulation visible at the wall-face site. **Cycle 3**: red (RED) shadow appears at the downstream site; sign flip across the bond identifies it in all three scans independently. Gold circles: bond endpoints. Stars: scan centers.

later at $t_1 = t_0+1$, the weaker bond slows drainage from u: weight accumulates, $n_{\text{pert}}(u) > n_{\text{base}}(u)$, so $\mathcal{L}(u) < 0$ (blue, wall face). Simultaneously, in the baseline weight flows through bond (u, v) and arrives at v, while in the perturbed run less weight crosses: $n_{\text{pert}}(v, t_1) < n_{\text{base}}(v, t_1)$, so $\mathcal{L}(v) > 0$ (red, shadow). The temporal order is universal: blue at the wall face first, red at the shadow one cycle later. The broken bond sits between the blue and red sites, identified by the sign change (Figure 1). The contrast $|\mathcal{L}(u) - \mathcal{L}(v)|$ is maximum at t_1, providing the detection signal.

Each center k is initialized and evolved *independently*: a separate baseline and perturbed simulation pair per center. The signed cross-entropy $\mathcal{L}_k(v)$ is computed for each center individually,

then summed into an ensemble log-likelihood:

$$\mathcal{L}_{\text{ens}}(v) = \sum_{k=1}^{p} \mathcal{L}_k(v) \tag{6}$$

Centers on the same side of the broken bond reinforce each other; centers on opposite sides contribute opposite signs that partially cancel. Silent centers (wave never reaches site v) contribute zero and do not corrupt the sum.

A more robust alternative to the pointwise $\mathcal{L}(v, t_0)$ accumulates the signal over all T cycles:

$$\mathcal{B}(v) = \sum_{t=1}^{T} n_{\text{base}}(v, t), \qquad \mathcal{N}(v) = \sum_{t=1}^{T} n_{\text{pert}}(v, t), \qquad \Lambda(v) = \ln \frac{\mathcal{B}(v)}{\mathcal{N}(v)} \tag{7}$$

Positive Λ identifies red (shadow), negative identifies blue (wall face). The integral captures both the delay and the sustained amplitude difference: a broken bond delays the perturbed wave *and* reduces its steady-state weight, making $\Lambda(v)$ more robust than $\mathcal{L}(v, t_0)$ when the pointwise signal is weak. In the multi-bond experiments, $\Lambda(v)$ is the primary detection signal.

5 The Log-Transfer Vector Field

The cross-entropy $\mathcal{L}(v, t) = \ln(n_{\text{base}}/n_{\text{pert}})$ is a *scalar field* on sites. We now show that its natural spatio-temporal gradient is a *vector field* defined on bonds, and that this gradient field directly reveals the broken bond as a physical singularity.

Since $n(v, t)$ is a scalar field on the graph, its discrete spatial gradient at site v in the direction of neighbor u is $(\nabla n)_{v\to u}(t) = n(u, t) - n(v, t)$. Since $n(v, t)$ is a probability, weight flows *multiplicatively* through bonds: $n(u, t+1) = n(v, t) \cdot T_{v\to u}$, so the natural gradient is in log-space. The *spatio-temporal gradient* of $\mathcal{L}$ at directed bond $(v \to u)$ from cycle t to $t+1$ is:

$$\begin{aligned}(\nabla_{\text{st}}\,\mathcal{L})_{v\to u}(t) \;&=\; \mathcal{L}(u,\, t+1) - \mathcal{L}(v,\, t) \\ &=\; \ln \frac{n_{\text{base}}(u, t+1)}{n_{\text{base}}(v, t)} - \ln \frac{n_{\text{pert}}(u, t+1)}{n_{\text{pert}}(v, t)} \;=\; \ln \frac{T^{\text{base}}}{T^{\text{pert}}} \;\equiv\; \delta \ln T_{v\to u}(t)\end{aligned} \tag{8}$$

At each site v, collecting this quantity over all neighbors $u_1, u_2, \ldots$ (up to 4 for a rectangular lattice) gives a *vector field*:

$$(\nabla_{\text{st}}\,\mathcal{L})_v(t) \;=\; \big[\,(\nabla_{\text{st}}\,\mathcal{L})_{v\to u_1}(t),\; (\nabla_{\text{st}}\,\mathcal{L})_{v\to u_2}(t),\; \ldots\,\big] \tag{9}$$

one component per bond direction. Each component is a log-likelihood ratio: positive when the baseline transfers more than the perturbed run along that bond, negative when the perturbed transfers more.

A broken bond (u, v) is a *singularity* in this gradient field. At the first wavefront crossing (cycle t), the baseline flows through the bond freely while the perturbed run is impeded: $\delta \ln T_{u\to v}(t) > 0$ (red, baseline transfers more through the broken bond). One cycle later, weight has accumulated at the wall-face site u in the perturbed run — $n_{\text{pert}}(u)$ grows, making the reverse ratio $n(v, t+1)/n_{\text{pert}}(u, t)$ larger: $\delta \ln T_{v\to u}(t) < 0$ (blue, perturbed transfers more in reverse because the shadow site v is depleted). The two directed components of the same bond have *opposite signs* — the gradient field has a discontinuity exactly at the broken bond.

By subsequent cycles the perturbation diffracts: the wall-face accumulation radiates excess weight along alternative paths, enriching bonds that would normally receive less, while the shadow

deepens. The gradient field develops a characteristic *diffraction pattern* centered on the broken bond — two diagonal fringes whose orientation encodes the bond direction directly (Figure 2).

The gradient field $\nabla_{\text{st}}\,\mathcal{L}$ has several properties that distinguish it from the scalar $\mathcal{L}(v,t)$. It is defined on *bonds* rather than sites, so the broken bond appears directly as a gradient discontinuity rather than through its effect on neighboring sites. It carries *directional* information — the sign asymmetry between forward and reverse components identifies which side is the wall face and which is the shadow, and the diffraction pattern orientation encodes the bond direction. The normalization by $n(v,t)$ suppresses the dominant-site bias that affects site-based metrics. Systematic artifacts from the simulation — Trotter color ordering, BP approximation, or any other effect present in both runs — cancel in the ratio $T_{\text{base}}/T_{\text{pert}}$, whether n_{base} and n_{pert} come from two simulations or from a simulation and hardware shots.

6 Bond Detection

For each bond b connecting sites v_a and v_b we define a *bond score*:

$$S(b) = \mathcal{L}_{\text{ens}}(v_a) \times \mathcal{L}_{\text{ens}}(v_b) \tag{10}$$

A negative product indicates a sign flip across the bond, the causal signature of the broken bond. Rank bonds by most negative $S(b)$.

Table 1 shows the individual and ensemble results for bond 12 (sites 15 $leftrightarrow$16, identity PTM perturbation, 7×7 lattice, $\chi = 50$, 5 cycles).

Center	Side	$\mathcal{L}(15)$	$\mathcal{L}(16)$	Result
10	upstream	+0.87	−0.02	✓CORRECT
30	upstream	+0.37	−0.04	✓CORRECT
25	downstream	+0.03	−0.04	× WRONG
32	downstream	+0.00	−0.03	× WRONG
6	too far	0	0	silent
34	too far	0	0	silent
Ensemble	—	+1.23	−0.13	✓CORRECT

Table 1: Signed cross-entropy at bond 12 endpoints (sites 15,16). Ensemble score: $S(12) = -0.164$, next bond $= -0.002$ (82× margin).

The ensemble identifies bond 12 with an 82-fold margin over the next candidate, despite only 2 of 6 individual centers giving the correct answer. The wrong individual answers: centers 25 and 32 both correctly identify site 16 as a bond endpoint; their error is only in declaring the wrong adjacent bond. The ensemble product cancels these partial errors.

7 Seismic Multi-Center Scanning

Rather than running each center independently, we initialize all p centers simultaneously in a single seismic scan: the initial operator is $O(0) = \sum_k Z_{c_k}$, firing all centers at once. At each site v, the first-arrival signal is dominated by the nearest center, automatically selecting the strongest directional component. The sign flip at the broken bond endpoints emerges without any explicit path tracing or per-center bookkeeping, and the cost drops from $2p$ simulations to just 2 (one baseline, one perturbed).

To achieve broad coverage with no geometric blind spots, we use three complementary center placements on the 7×7 lattice: corners and center ($\{1, 7, 25, 43, 49\}$), quadrant centers ($\{11, 13, 37, 39\}$), and mid-edge sites ($\{4, 22, 28, 46\}$). Each placement covers a different region of the grid; together the three scans ensure every bond is within $d_{\max} = 2$ hops of at least one center in at least one scan. At $v_{\text{hop}} \approx 0.45$ hops/cycle, 2 hops requires 5 simulation cycles. The total cost is 6 simulations (3 baselines + 3 perturbed), run in parallel on 3 GPUs.

Bond scores are computed using the integrated log-ratio $\Lambda(v)$ across all three scans. For each bond b connecting sites v_a and v_b, the score is the maximum contrast $|\Lambda(v_a) - \Lambda(v_b)|$ over all scans. The top-k bonds by score are declared broken, where k is the number of expected faults. On 20 single-bond trials ($\delta J = -0.20$, two independent datasets), this procedure achieves **20/20** correct identifications. On 10 two-bond trials it achieves **9/10**, with the single failure corresponding to two bonds sharing an endpoint whose wavefronts cancel (the BLUE-BLUE-RED interference pattern discussed in Section 10).

8 Tensor Network Simulation as Quantum Circuit Execution

The Trotter simulation underlying our method is not merely an approximation of a quantum circuit; it *is* the quantum circuit, represented in the Pauli operator basis rather than the amplitude basis. The identification is exact:

- Each **lattice site** = one qubit
- Each **bond gate** $\exp(-i\theta(XX + YY + ZZ))$ = one two-qubit gate
- Each **color class** (non-overlapping bonds applied in parallel) = one gate layer
- Each **belief propagation synchronization** after a full Trotter cycle = one barrier (per-color-class BP, as in the Julia L-MPF implementation [3], provides finer sub-cycle resolution matching the quantum circuit barrier structure)
- Each **PAULI_NL output** = a mid-circuit Pauli weight measurement

Both the graph coloring (determining which bonds share qubits) and the quantum circuit parallelism constraint (which gates can be applied simultaneously) derive from the lattice topology. The color class ordering is the gate layer ordering.

This identification reveals two distinct timescales. *Physical time* t counts integer Trotter cycles, $t = n \cdot \varepsilon$; the observables $C(t)$ and $n(v, t)$ at cycle granularity are physical, independent of color class ordering to $O(\varepsilon^2)$ and convergent to true quantum dynamics. *Circuit time* t_{color} has sub-cycle, per-color-class granularity, a property of the circuit implementation rather than the Hamiltonian. Two Trotter implementations of the same Hamiltonian with different color orderings agree on $C(t)$ to first order in ε but differ by $O(\varepsilon^2)$ Trotter errors; their $n(v, t_{\text{color}})$ tomography fingerprints differ already at the first sub-cycle. The BP synchronization granularity determines which timescale is accessible: per-cycle BP (as used here) loses sub-step information; per-color BP (Julia L-MPF [3]) respects the causal barrier structure of the quantum circuit.

A perturbed bond (coupling $J \neq 1$) is a *miscalibrated gate*: the two-qubit rotation angle $\theta = J \cdot \varepsilon$ differs from the nominal value. The Pauli transfer matrix for $\exp(-i\theta(XX+YY+ZZ))$ has diagonal entries $\cos(\theta)$, which for any $J \neq 1$ gives $\cos(J\varepsilon) \neq \cos(\varepsilon)$. Since $|\cos(\theta)| \leq 1$ for all θ (unitary gate), any coupling deviation attenuates the wave relative to baseline, guaranteeing $n_{\text{pert}}(v^*) < n_{\text{base}}(v^*)$ at the shadow endpoint v^*, so $\mathcal{L}(v^*) > 0$ (red) regardless of the sign of $\delta J = J - 1$. The detection method is robust: the red site is always the shadow endpoint, never inverted.

This frames our tomography as *quantum ATPG* (Automatic Test Pattern Generation) [4], the classical circuit testing methodology generalized to quantum gates. In classical ATPG: fault activation (excite the stuck-at gate), path sensitization (find an observable path from fault to output), backward implication (infer the faulty gate). Here: perturbation (change J), wavefront propagation (operator spreading from center to shadow site), and cross-entropy inversion (red site + center position $\rightarrow$ bond identity).

9 Discussion

The observable $n(v,t)$ is accessible on real quantum hardware via local Pauli measurements after ZNE gate-folding circuits [1]. The cross-entropy $\mathcal{L}(v)$ requires only the ratio of two measurements, with no absolute calibration. The center placement is determined by the graph topology alone, independent of the Hamiltonian.

The method exploits the Lieb-Robinson bound: information cannot propagate faster than v_{hop} hops/cycle. A broken bond creates a causal anomaly; the wavefront on one side cannot inform the other. The first-arrival time $t_{\text{first}}(v)$ is observed directly, without knowing v_{hop} in advance.

The seismic 3-scan cost is fixed at 6 simulations regardless of how many bonds are broken. On the 7×7 lattice ($\chi = 50$, 5 cycles, AMD MI100 32 GB GPU), each scan pair runs in 272 s using 11% VRAM with 2 GPUs in parallel.

When two adjacent broken bonds share an endpoint, say bonds (i,j) and (j,k) with shared site j, the cross-entropy field shows a three-site pattern: site i appears red (shadow of bond (i,j)), site k appears blue (wall face of bond (j,k)), and the shared site j appears white. At j, the red contribution from one broken bond and the blue contribution from the other cancel, producing an *interference node*. The white fringe is not a detection failure; it is an interference pattern arising from opposite-side contributions canceling at the shared node. The current algorithm detects such pairs via a triplet scorer $S_{\text{wall}} = \mathcal{L}(v_i) \times \mathcal{L}(v_k)$, skipping the white shared endpoint and reading the outer sites directly.

The seismic and aperture methods are naturally complementary and suggest a two-phase adaptive protocol. In **Phase 1** (seismic), all p centers fire simultaneously: one quantum experiment, minimum cost, producing a map of RED sites that identifies the region of interest. In **Phase 2** (aperture), the RED sites from Phase 1 guide the selection of targeted observation centers geometrically positioned to discriminate between candidate bonds adjacent to those sites. Only $k \ll p$ centers need to be run, each independently.

This mirrors standard practice in other imaging domains: a wide-field scout scan (seismic) followed by targeted high-resolution acquisition (aperture). On quantum hardware, Phase 1 maps directly to a single quantum experiment; Phase 2 selects the minimum number of additional experiments to confirm the bond identity.

The seismic phase is the most natural starting point: it requires only one circuit (all centers in a single initialization) and one set of hardware shots for the perturbed device. The baseline $n_{\text{base}}(v,t)$ is provided by simulation and reused across all candidate perturbations at no additional hardware cost. The aperture phase then tests the hypothesis with directional projections; perturbation signals that survive across independent projections with geometry-consistent timing are physical, while chi-truncation artifacts and shot noise are uncorrelated across runs and average out.

Wavefront triangulation across scans. When multiple independent scans are available, the temporal order in which each scan detects the broken bond carries geometric information beyond

the sign flip alone. This enables a triangulation strategy analogous to GPS: each scan provides a time-of-arrival constraint that localizes the bond.

Formally, scan s with nearest center c_s detects the broken bond at cycle t_s^*, the first cycle where $|\mathcal{L}_k(v)| > \varepsilon_{\text{th}}$ for some bond endpoint v. This constrains the bond to lie within $d_s = t_s^* \times v_{\text{hop}}$ hops of c_s: a ball $\mathcal{D}(c_s, d_s)$ on the graph. Across M scans, the bond lies in the intersection:

$$b^* \in \bigcap_{s=1}^{M} \mathcal{D}(c_s,\, t_s^* \times v_{\text{hop}}) \tag{11}$$

where $\mathcal{D}(c, r)$ is the set of bonds within crossing distance r of center c.

The sign of the earliest detection (which endpoint is RED vs BLUE) further constrains the direction: the center c_s is on the BLUE (wall-face) side of the broken bond. Subsequent scans confirm or deny: a scan detecting the *opposite* sign at the same endpoints confirms (center is on the other side of the same bond); a scan detecting different endpoints shifts the candidate set. Each confirmation tightens the intersection; each denial eliminates candidates.

In practice on the 7×7 lattice with 3 scans (corners+center, quadrants, mid-edges): scan 2 detects bond 9 (11↔12) at cycle 1 (center 13, 1 hop away), scan 3 confirms with opposite sign at cycle 2 (center 4, 2 hops away), scan 1 provides weak confirmation at cycle 2 from the far corners. The intersection of the three balls gives a small candidate set containing the true bond.

BFS-guided fault detection algorithm

The detection method reduces to a graph-theoretic algorithm that operates entirely on the observable $n(v, t)$, without reference to the Hamiltonian, coupling values, or simulation parameters. Let $G = (V, E)$ be the coupling graph, $\{c_s\}$ the observation centers, and ε_{th} the detection threshold.

1. For each center c_s and each site $v \in V$, record the baseline arrival time $t^*_{\text{base}}(v) = \min\{t : n_{\text{base}}(v, t) > \varepsilon_{\text{th}}\}$ and perturbed arrival time $t^*_{\text{pert}}(v) = \min\{t : n_{\text{pert}}(v, t) > \varepsilon_{\text{th}}\}$. These are the actual BFS frontier crossings, not predicted hop distances; $t^*_{\text{pert}}(v)$ may exceed $t^*_{\text{base}}(v)$ if a broken bond delays the perturbed wave.

2. Both the simulation and the quantum circuit assume an execution order among independent bonds, determined by Trotter color class. We materialize this order for causal determination: assign each site v the pair $(h(v), c(e_v))$ where $h(v)$ is its BFS hop depth and $c(e_v) \in \{0, 1, 2, 3\}$ is the color class of the bond used to reach it. The total order (h, c) gives exact sub-cycle causality: $(h_u, c_u) < (h_v, c_v)$ means site u is strictly upstream of v.

3. For a bond (u, v) with u upstream of v, compute the bond transfer coefficient:

$$T = \frac{n(v,\, t^*_{\text{base}}(v))}{n(u,\, t^*_{\text{base}}(u))} \tag{12}$$

   When $n_{\text{base}}(u) \gg \varepsilon_{\text{th}}$, both waves share the same source amplitude at the BFS frontier and $\mathcal{L}(v) \approx \ln(T_{\text{base}}/T_{\text{pert}})$: the signed cross-entropy directly measures the relative change in bond transfer. A broken bond reduces $T_{\text{pert}} < T_{\text{base}}$, giving $\mathcal{L}(v) > 0$ (red shadow) and $\mathcal{L}(u) < 0$ (blue wall face).

4. For each site v with $|\mathcal{L}(v)| > \varepsilon_{\text{th}}$, examine neighbors u with $(h_u, c_u) < (h_v, c_v)$. Declare bond (u, v) as a candidate if $\mathcal{L}(u)$ and $\mathcal{L}(v)$ have opposite signs. Score each candidate by $|\mathcal{L}(u) - \mathcal{L}(v)| \cdot \delta/t^*$ where $\delta = 2$ if $t^*_{\text{pert}}(v) > t^*_{\text{base}}(v)$, else 1, and $t^* = \min(t^*_{\text{base}}(u), t^*_{\text{base}}(v))$.

5. Repeat steps 1–4 for each scan independently. Rank all candidate bonds by total score across scans and declare the top-k as broken.

From our experiments we have identified three recurrent interference signatures, which we name here as a working vocabulary based on observed cases. Each is a consequence of the BFS frontier geometry and the Trotter color class ordering, with no quantum-specific reasoning required. We expect further patterns to emerge as the geometry of faults and center placements varies; the vocabulary below reflects our current observations, not an exhaustive classification. **RED-BLUE** is the clean single-bond signature: center on one side, sign flip across the bond. **BLUE-WHITE-BLUE** arises when a bond is sandwiched between two equidistant centers: the shared site is white (interference node) and the outer sites are blue on both sides, making both adjacent bonds identifiable. **BLUE-BLUE-RED** occurs when two orthogonal broken bonds share an endpoint: the shared site accumulates from two directions, the outer RED identifies one bond, and the adjacent BLUEs identify the second.

Experimental results. On a 7×7 lattice with 3 scans (13 centers, 3 GPUs parallel, $d_{\max} = 2$), testing 10 random trials with 2 simultaneous broken bonds ($\delta J = -0.20$): the BFS algorithm with sub-cycle color-class ordering finds **both bonds correctly in 9/10 trials** (90%), with 1 partial case and 0 complete failures. The single partial failure (trial 5) corresponds to two orthogonal bonds sharing an endpoint where both centers' waves are simultaneously blocked: the BLUE−BLUE−RED pattern, representing the hardest geometric configuration.

Failure taxonomy. We characterize five failure modes of increasing complexity:

1. **Multiple random bonds**: shadows may overlap, making the $\mathcal{L}$ field ambiguous
2. **Collinear bonds**: a line of failures looks like a thick wall; bond resolution degrades
3. **Hexagonal cluster**: a faulty qubit touches 2–3 bonds simultaneously, producing a RED cluster with BLUE halo
4. **Graduated failure** (δJ smooth variation): weak signal approaching the sensitivity limit

These represent the primary open experimental questions.

Limitations. The current experiment uses the identity PTM, a maximally disruptive perturbation that is deliberately clean. Cases 2-5 above are designed to stress-test the method: case 2 tests whether overlapping shadows can be resolved, case 3 tests spatial resolution limits, case 4 tests cluster detection on irregular topologies, and case 5 tests the sensitivity threshold for realistic coupling deviations.

10 Multi-Bond Detection and Iterative Calibration

We extend the seismic BFS algorithm to detect k simultaneously broken bonds by declaring the top-k candidates from the bond score ranking (Table 2).

The per-bond detection rate remains above 90% even at 8 simultaneous faults. Systematic failures are geometrically localized: bond 8 (sites 10↔11) and bond 36 (sites 43↔44) consistently score below the top-k threshold due to weak seismic illumination from the three scan center placements. These are not random failures; they reflect the coverage geometry of the 3-scan plan.

k bonds	Trials	Bonds found	Per-bond rate
1	20	20/20	100%
2	10	19/20	95%
4	5	18/20	90%
8	2	15/16	94%

Table 2: Per-bond detection rate (single seismic iteration, top-k declaration, $\delta J = -0.20$, 7×7 lattice, $\chi = 50$, 5 cycles). The per-bond detection rate remains above 90% even at 8 simultaneous faults.

The detection naturally supports an iterative calibration loop. One seismic pass finds above 90% of the broken bonds; the hardware is recalibrated for those; a second seismic pass then searches the residual, where the remaining faults now surface with a cleaner signal unclouded by the bonds already fixed. The process repeats until $n_{\text{pert}}(v,t) \approx n_{\text{base}}(v,t)$ everywhere. At 8 simultaneous faults, 7 are found in the first pass; fixing those makes the last one detectable at 100% accuracy in the second. The stopping criterion is physical and requires no additional measurement overhead.

The iterative structure also justifies the design choices. Short cycles (3–5) are sufficient because only the next bond to fix needs to be ranked, not all faults simultaneously. A moderate threshold $\varepsilon_{\text{th}} = 10^{-3}$ maximizes BFS coverage so no bond is in a blind spot. Three seismic placements cover all 84 bonds with no geometric dead zones except the small set of bonds with weak illumination from all three centers, exactly the bonds that surface cleanly in iteration 2 once their neighbors have been fixed, with no additional simulations required.

11 Measurability and Detection Thresholds

The detection method requires $n(v,t)$ to be measurable both from the simulation fidelity standpoint and from the hardware shot budget. Two independent conditions must hold simultaneously.

Recall that $n(v,t) \in [0,1]$ is a probability: the fraction of Pauli basis operators that have non-identity support at site v after t Trotter cycles (Section 3). On real hardware the circuit is executed N times. A barrier after each Trotter cycle allows a mid-circuit local Pauli weight measurement at site v; each shot returns either the identity at v or a non-identity Pauli (X, Y, or Z). The three non-identity outcomes are not distinguished — only the binary question "identity or not" is asked, and the non-identity count k accumulates as a success count. This binary nature of each shot makes the Binomial model natural: $k \sim \text{Binomial}(N, n(v,t))$, where the physical probability $n(v,t)$ is the Bernoulli success probability for each of the N independent shots. The empirical estimate $\hat{n}(v,t) = k/N$ is then an unbiased estimator of the true $n(v,t)$.

The estimator has variance $\text{Var}(\hat{n}) = n(1-n)/N$, a direct consequence of the Binomial distribution. At the frontier, $n(v,t) \ll 1$: the total Pauli weight is conserved ($\sum_v n(v,t) = 1$) and spreads over a growing shell of sites, so each individual $n(v,t)$ decreases with distance from the center. In this small-n regime $1-n \approx 1$ and the variance simplifies to n/N, giving a shot noise floor $\sigma(n) \approx \sqrt{n/N}$. Note also that only sites where $n(v,t) > \varepsilon_{\text{th}}$ contribute to the detection (Section 3), so the relevant regime is $\varepsilon_{\text{th}} \leq n(v,t) \ll 1$. The baseline $n_{\text{base}}(v,t)$ is provided by simulation and is exact; only the perturbed run is measured on hardware. The condition is that $\hat{n}_{\text{pert}}$ is a reliable estimate of the true $n_{\text{pert}}(v,t)$, so that the comparison with the theoretical $n_{\text{base}}(v,t)$ is meaningful. Reliability requires the standard deviation of the estimator $\sigma(\hat{n}) = \sqrt{n/N}$ to be much smaller than the quantity being estimated: $\sigma(\hat{n}) \ll n$. As N grows, $\sigma(\hat{n})$ shrinks and the estimate converges to the true value. The minimum observable n is set by the detection threshold ε_{th}. For $\hat{n}$ to be a

reliable estimate at $n = \varepsilon_{\text{th}}$ we require $\sigma(\hat{n}) \leq \varepsilon_{\text{th}}$, i.e., $\sqrt{\varepsilon_{\text{th}}/N} \leq \varepsilon_{\text{th}}$, which gives:

$$N \geq \frac{1}{\varepsilon_{\text{th}}} \tag{13}$$

For $\varepsilon_{\text{th}} = 10^{-3}$, at least 10^3 shots are required. Table 3 shows the statistical floor as a function of shot budget.

N shots	$\sigma(\hat{n})$ at $n = \varepsilon_{\text{th}}$	Status
10^2	3×10^{-3}	noise exceeds threshold
10^3	1×10^{-3}	noise matches threshold (minimum)
10^4	3×10^{-4}	reliable estimate
10^6	3×10^{-5}	well into reliable regime

Table 3: Estimator noise $\sigma(\hat{n}) = \sqrt{\varepsilon_{\text{th}}/N}$ at the detection threshold $\varepsilon_{\text{th}} = 10^{-3}$. Reliable detection requires $N \geq 1/\varepsilon_{\text{th}} = 10^3$.

The tensor network simulation introduces two distinct noise sources that set a floor on trustworthy $n(v,t)$ values. First, truncation: bond singular values below the cutoff δ_{cut} are set to zero, so any $n(v,t)$ arising from truncated components is numerical noise, not physical operator spreading. Second, belief propagation is not strictly causal: messages propagate globally in each sweep, producing tiny non-zero $n(v,t) \sim 10^{-4}$ at sites outside the Lieb-Robinson lightcone. In a real quantum circuit these sites would never be activated — only gates within the lightcone fire — so this leakage is a simulation artifact with no hardware counterpart. Both effects are below $\delta_{\text{cut}} = 10^{-3}$, which we therefore adopt as the simulation floor.

Other simulators with tighter cutoffs could in principle trust smaller $n(v,t)$ values and use a lower ε_{th}. However, a smaller threshold demands proportionally more shots at the frontier: to estimate $n \approx \delta_{\text{cut}}$ reliably one needs $N \geq 1/\delta_{\text{cut}}$ shots. A simulation floor of 10^{-6} would require 10^6 shots merely to reach reliable estimates at the threshold — a cost that must be weighed against the physical benefit of the finer resolution.

The two conditions converge at the same threshold $\varepsilon_{\text{th}} = \delta_{\text{cut}} = 10^{-3}$: the simulation floor sets the minimum trustworthy signal, and from Table 3 reliable hardware estimates at that level require $N \geq 1/\varepsilon_{\text{th}} = 10^3$ shots. Both conditions are satisfied simultaneously by:

$$\varepsilon_{\text{th}} \geq \delta_{\text{cut}}, \qquad N \geq \frac{1}{\varepsilon_{\text{th}}} \tag{14}$$

A tighter simulation cutoff $\delta_{\text{cut}} < 10^{-3}$ enables a smaller ε_{th} and finer resolution, but demands proportionally more shots at the frontier to maintain reliable estimates.

From Section 8, the bond gate PTM has diagonal entries $\cos(J\varepsilon)$. A coupling deviation δJ shifts the rotation angle and attenuates the diagonal entry by $-\sin(J\varepsilon) \cdot \varepsilon\, \delta J$ to first order. At $J = 1$ this means the perturbed weight at site v is simply the baseline attenuated:

$$n_{\text{pert}}(v,t) \approx n_{\text{base}}(v,t) \left[1 - \varepsilon \sin(\varepsilon) |\delta J|\right] \tag{15}$$

so the signal $\delta n = n_{\text{base}} - n_{\text{pert}} \approx n_{\text{base}}(v,t) \cdot \varepsilon \cdot \sin(\varepsilon) \cdot |\delta J|$ is proportional to the baseline weight and scales linearly with the coupling deviation. The attenuation factor $\varepsilon \sin(\varepsilon) |\delta J|$ depends only on the Trotter step and the perturbation magnitude, not on the site.

Note that this is a stronger requirement than simply estimating $\hat{n}_{\text{pert}}$ reliably: we need the signal δn to be measurably larger than the estimator noise $\sigma(\hat{n}) \approx \sqrt{n_{\text{base}}/N}$. Setting $\delta n \geq \sigma(\hat{n})$

gives the minimum detectable coupling deviation. We emphasize that the present work performs *detection* — identifying the location and existence of a broken bond from wavemap patterns — not *estimation* of δJ from the numerical values of $n(v,t)$. Quantitative estimation of the coupling deviation from the measured $\hat{n}_{\text{pert}}$ is a harder and separate problem, left for future work.

$$|\delta J|_{\min} \approx \frac{\varepsilon_{\text{th}}}{\varepsilon \cdot \sin(\varepsilon) \cdot \sqrt{N \cdot n_{\text{base}}(v,t)}} \tag{16}$$

For $\varepsilon = 0.10$ and $n_{\text{base}} \approx 10^{-2}$ (genuine wavefront at first arrival), the detection threshold scales as $|\delta J|_{\min} \propto 1/\sqrt{N}$: at $N = 10^6$ shots $|\delta J|_{\min} \approx 1.0$, at $N = 10^7$ it drops to 0.32. To detect the experimental coupling degradation $|\delta J| = 0.20$ used throughout this work requires $N \approx 2.5 \times 10^7$ shots. This is independent of the minimum shot budget for reliable estimation ($N \geq 10^3$): the two conditions address different questions : can we measure $n(v,t)$ at all, versus can we resolve the specific coupling deviation of interest.

12 Conclusion

A broken bond leaves a causal signature in the operator wavefront: a red-blue sign flip in $\mathcal{L}(v,t)$ at its two endpoints, identifiable from the first wavefront crossing and robust across all center placements. Three seismic scans, each firing a different set of centers simultaneously, cover all 84 bonds of the 7×7 lattice in 6 simulations and achieve 20/20 single-bond detection. The integrated log-ratio $\Lambda(v)$ extends this to realistic coupling degradation ($\delta J = -0.20$) and multiple simultaneous faults, maintaining a per-bond detection rate above 90% for up to 8 faults in a single pass. The iterative calibration loop then reduces any multi-bond problem to a sequence of single-bond problems, each solved at 100% accuracy, with wavefront convergence as the natural stopping criterion.

The measurability analysis shows that $N \geq 10^3$ shots suffice for reliable estimation of $n(v,t)$ at the detection threshold; resolving the specific coupling deviation $|\delta J| = 0.20$ requires $N \sim 2.5 \times 10^7$ shots, a stronger condition that governs hardware deployment.

The method is a direct sequitur to the Lindblad MPF framework. Both use the same observable $n(v,t)$, the same circuits, and the same hardware protocol; MPF infers noise from a known Hamiltonian, while this work infers coupling geometry from a known noise baseline. Two inferences from one measurement, in opposite directions.

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A Aperture Synthesis with Voronoi Center Placement

An alternative to the seismic approach runs each center independently, in direct analogy with aperture synthesis in radio interferometry. Each center k at a different graph position plays the role of one antenna at a different baseline; $\mathcal{L}_k(v)$ encodes which side of the broken bond site v is on (sign) and the signal strength (magnitude).

Centers are placed with the constraint that no two share the same row or column in the lattice. This guarantees every bond has a strictly unique nearest center (no Voronoi ties): the perpendicular bisector between any two centers never passes through a bond, so one center always dominates the signal at every bond. For $p = 7$ on the 7×7 lattice (one center per row, one per column, a permutation placement), exhaustive search over permutations minimizing $\max_b T_b$ gives centers $\{3, 13, 15, 25, 35, 37, 47\}$ with zero ties across all 84 bonds, maximum $T_b = 5$ cycles, and balanced Voronoi regions of 10–16 bonds each.

Ensemble summation $\mathcal{L}_{\text{ens}}(v) = \sum_k \mathcal{L}_k(v)$ fails for bonds sandwiched between centers on opposite sides, where RED and BLUE contributions partially cancel. A *winner-takes-all* rule avoids this: for each center k compute the signed contrast across every bond,

$$C_k(b) = \left| \mathcal{L}_k^{\text{signed}}(v_a) - \mathcal{L}_k^{\text{signed}}(v_b) \right| \tag{17}$$

and declare the globally winning center-bond pair $(k^*, b^*) = \arg\max_{k,b} C_k(b)$. The winning center is necessarily on the BLUE side of the broken bond; each center is a fully independent observer and the one with the strongest signal wins. On 10 trials ($\delta J = -0.20$), the adaptive seismic-then-aperture protocol achieves **10/10** correct identifications: 7 bonds identified by seismic alone (2 simulations), 3 by aperture (14 simulations), at an average cost of $2 + 0.3 \times 14 \approx 6$ simulations per bond.

The seismic 3-scan approach in Section 6 supersedes this method for the multi-bond case: it achieves the same single-bond accuracy at lower cost, and extends naturally to multiple simultaneous faults without requiring the no-shared-row-or-column placement constraint.

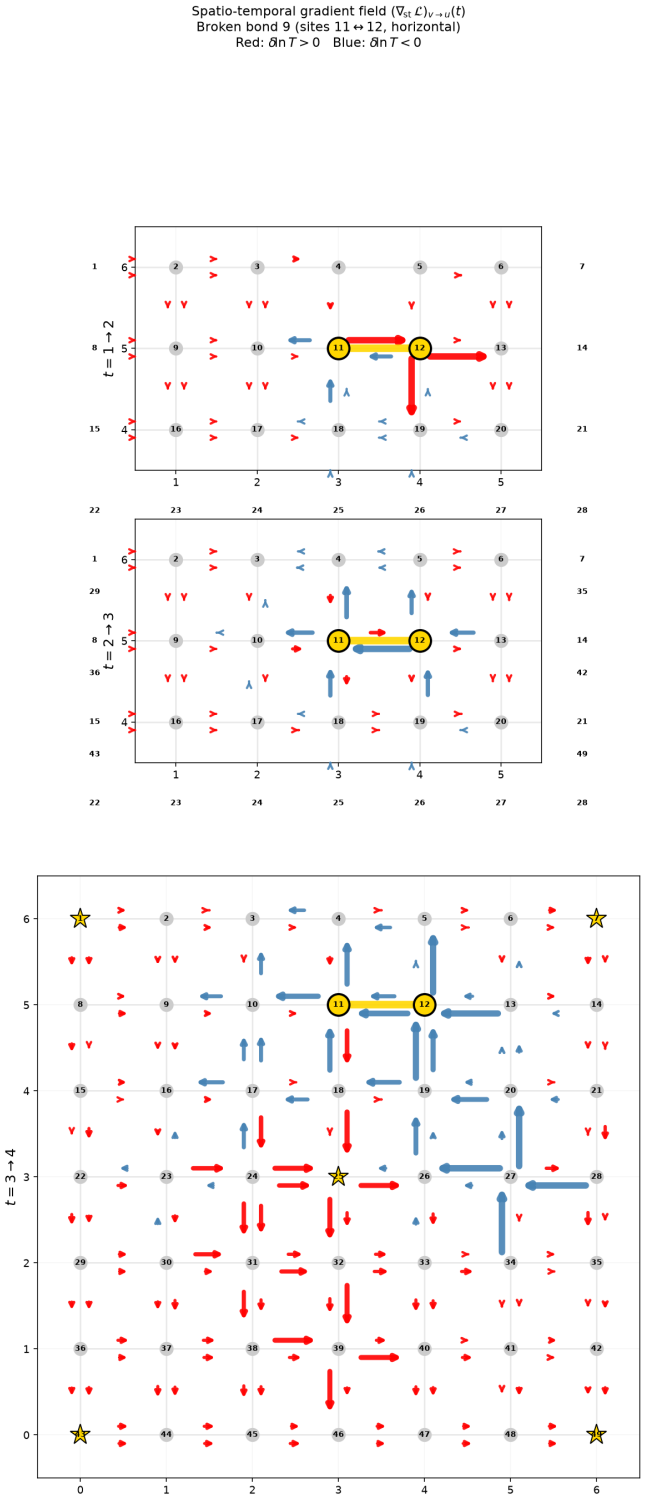


Figure 2: Spatio-temporal gradient field $(\nabla_{\mathrm{st}}\,\mathcal{L})_{v\to u}(t)$ for broken bond 9 (sites 11↔12, horizontal), rect 7×7, $\chi = 50$, scan 1 (corners+center). Each row shows one cycle transition. Red arrows: $\delta \ln T > 0$ (baseline transfers more through the bond). Blue arrows: $\delta \ln T < 0$ (perturbed transfers more in reverse). Gold bar: broken bond. Gold circles: bond endpoints. Gold stars: scan centers. **Top** ($t = 1 \to 2$): the wavefront first crosses the broken bond; large red arrow at bond 11→12 is the primary signal. **Middle** ($t = 2 \to 3$): wall-face accumulation at site 11 reverses the dominant direction; blue arrows appear as the perturbed run now transfers more from the enriched wall face. **Bottom** ($t = 3 \to 4$): the diffraction pattern is fully developed — a blue diagonal (shadow path) and a red diagonal (rerouted path) emerge, pointing at the broken bond and encoding its orientation.